Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 28: Infinitely Many Cases, Two Steps

The Principle of Mathematical Induction

Chartrand & Zhang, Section 4.1, pp. 115–123

Name: _____ Date: _____

The question this course has been asking since Week 1

In Homework 1 you checked $n^2 - n + 41$ for forty values and watched it fail at the forty-first. The lesson, repeated all term: **checking cases cannot prove a claim about infinitely many integers**. You would never finish.
But in Homework 2 Problem 7 you also saw that checking *every* case *is* a proof — when there are finitely many. Today's method resolves the tension: it proves infinitely many statements with an argument of exactly two steps.

By the end of today you should be able to

1. State the Principle of Mathematical Induction.
2. Write an induction proof with a correct base step and inductive step.
3. Explain why an induction proof with no base step proves nothing.

1. The principle

The Principle of Mathematical Induction

The statement $\forall n \in \mathbb{N},\ P(n)$ is true if
(1) $P(1)$ is true, *(base step)*
(2) $\forall k \in \mathbb{N},\ P(k) \Rightarrow P(k+1)$ is true. *(inductive step)*

Why it works: (1) gives $P(1)$. Then (2) with $k = 1$ gives $P(1) \Rightarrow P(2)$, so $P(2)$ holds by modus ponens. Then $k = 2$ gives $P(3)$, and so on without end.

But "and so on" is not a proof

That chain is the intuition, not the justification — you can never finish writing it out. Chartrand is explicit: the Principle is taken as an **axiom** about the positive integers, accepted without proof. What *you* prove is steps (1) and (2).

Look at what step (2) is

Step (2) is a universal implication, $\forall k,\ P(k) \Rightarrow P(k+1)$. You prove it with a **direct proof**: assume $P(k)$ for an arbitrary k, and derive $P(k+1)$.
So every induction proof contains a Chapter 3 proof inside it. The assumption $P(k)$ is called the **inductive hypothesis**.

2. The template

Every induction proof has four parts

1. **Announce it:** "We proceed by induction."
2. **Base step:** verify $P(1)$ directly.
3. **Inductive step:** "Assume $P(k)$ for a positive integer k. We show $P(k+1)$." Then prove it — and *say where you use* the inductive hypothesis.
4. **Conclude:** "By the Principle of Mathematical Induction, $P(n)$ holds for every positive integer n."

3. Worked: Result 4.4

Look first: $1 = 1$, $1+3 = 4$, $1+3+5 = 9$, $1+3+5+7 = 16$.

Result 4.4

For every positive integer n, $\quad 1+3+5+\cdots+(2n-1) = n^2$.

Proof. We proceed by induction.

Base step. For $n = 1$ the left side is 1 and the right side is 1^2. They agree.

Inductive step. Assume that

$$1+3+\cdots+(2k-1) = k^2$$

for a positive integer k. We show $1+3+\cdots+(2k+1) = (k+1)^2$. Split off the last term:

$$\begin{aligned} 1+3+\cdots+(2k-1)+(2k+1) &= \big[1+3+\cdots+(2k-1)\big]+(2k+1) \\ &= k^2+(2k+1) && \text{by the inductive hypothesis} \\ &= (k+1)^2. \end{aligned}$$

By the Principle of Mathematical Induction, the formula holds for every positive integer n. ■

The move that makes every sum proof work

Split off the last term. The sum up to $k+1$ is the sum up to k, plus one more term. The sum up to k is exactly what the inductive hypothesis tells you about — substitute it.
If you never replace anything using the hypothesis, you have not done an induction proof.

4. The warning

Here is a "proof" that $3n^2 - n + 3$ is even for every positive integer n.

An inductive step that checks out

Assume $3k^2 - k + 3$ is even, say $= 2\ell$. Then

$$3(k+1)^2 - (k+1) + 3 = (3k^2 - k + 3) + 6k + 2 = 2\ell + 2(3k+1) = 2(\ell + 3k + 1),$$

which is even. *The algebra is completely correct.*

But at $n = 1$: $3(1)^2 - 1 + 3 = 5$, which is odd. In fact $3n^2 - n + 3$ is **never** even.

Why the step "worked"

The inductive step proves $P(k) \Rightarrow P(k+1)$. Since $P(k)$ is *never true*, every one of those implications has a false hypothesis — they are all vacuously true.
That is Lecture 4, exposing a broken proof. An inductive step with no base step can be true for a claim that is false everywhere. **The base step is not a formality**; it is what connects the chain to something real.

Exit ticket

1. Write the base step and the inductive hypothesis for proving $2 + 4 + \cdots + 2n = n(n+1)$.

2. Why does an inductive step alone not prove anything?

Before Wednesday

Read Section 4.2, "Additional Examples of Induction Proofs," pp. 123–128.

Self-check (not collected). Section 4.1, Exercises 1, 3, 5, and 9 — odd-numbered, answers in the back.

Coming up

Homework 10 is due Friday, October 30, 11:59 p.m.
Quiz 9 is Monday, November 2 — Sections 4.1 and 4.2.

Questions I still have